

Problem Sheet 4 Solutions

1.

The feedback network for the inverting amplifier is shown in the figure. V_A is the voltage at the inverting '-' input and Rule I says this is 0 V. Rule II says that whilst V_A is at 0 V, it is not *connected to ground*. No current flows into the input. This is what is known as a virtual ground. We can write two equations for the potential differences across the resistors:

$$\begin{aligned} V_{in} - I_1 R_{in} &= 0 \\ 0 - I_2 R_f &= V_{out} \end{aligned}$$

Rule II tells us $I_1 = I_2$, so

$$\begin{aligned} \frac{V_{in}}{R_{in}} &= -\frac{V_{out}}{R_f} \\ \text{Gain} = \frac{V_{out}}{V_{in}} &= -\frac{R_f}{R_{in}}. \end{aligned}$$

2.

(a) The RC input stage is a high-pass filter so we call this circuit an AC-coupled non-inverting amplifier. It rejects DC and low-frequency signals, and so would be used in situations where you have either a small AC signal superimposed on a large DC background or where a high-frequency signal had a significant low-frequency noise component which we wish to reject. The capacitor is often known as a 'blocking' or 'de-coupling' capacitor.

(b) The cut-off frequency for the input filter is 16 Hz so at frequencies well above this the filter has no effect

$$\text{Gain} = 1 + \frac{R_f}{R_0} = 51$$

At low frequencies, the RC filter looks like a voltage divider, with one fixed resistor $R_{in} = 100 \text{ k}\Omega$ and one frequency-dependent impedance $Z_c = 1/j\omega C$, where $\omega = 2\pi f$.

The voltage at V_{in+} is thus

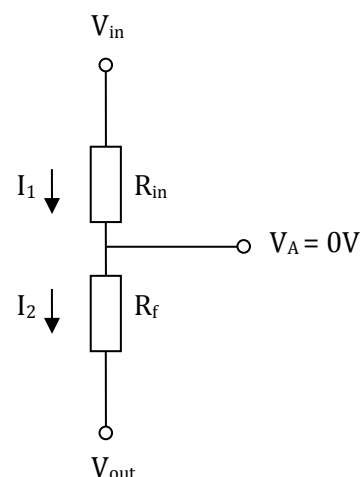
$$V_{in+} = \frac{V_{in} R_{in}}{|R_{in} + Z_c|} = \frac{V_{in} R_{in}}{\sqrt{R_{in}^2 + \frac{1}{\omega^2 C^2}}}$$

For $f = 10 \text{ Hz}$

$$V_{in+} = 0.53 V_{in}$$

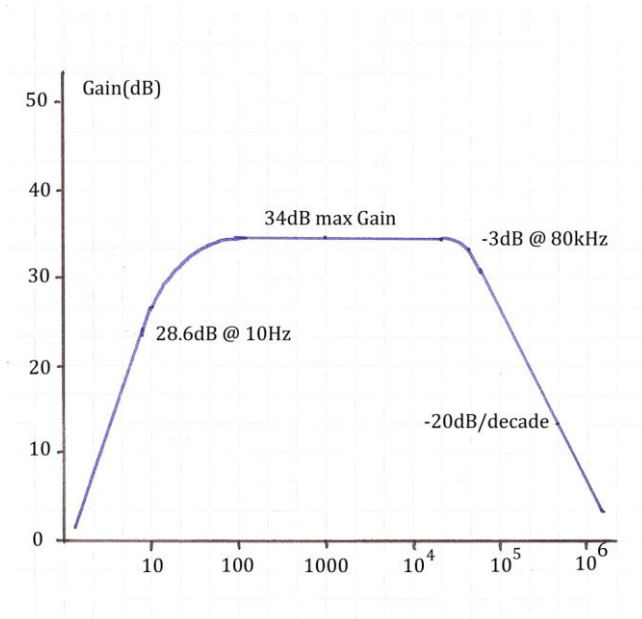
So net gain is

$$51 \times 0.53 = 27$$



Instrumentation

(c) Since $Gain_{dB} = 20 \log_{10} \frac{V_{out}}{V_{in}}$ we can expect a gain vs. frequency behaviour similar to the sketch shown in the figure overleaf.



3.

(a) R_f and R_0 form a potential divider from V_{out} to ground, with the inverting input of the op-amp at the centre of the divider. If we define the voltage at the inverting input to be V_A then

$$V_A = \frac{V_{out} R_0}{R_f + R_0}$$

β is defined as the fraction V_A/V_{out} so

$$\beta = \frac{R_0}{R_f + R_0}$$

(b) Rule 1 says $V_A = V_{in}$ so

$$V_{in} = \frac{V_{out} R_0}{R_f + R_0}$$

$$Gain = \frac{V_{out}}{V_{in}} = 1/\beta$$

(c) $V_A \neq V_{in}$, however,

$$\begin{aligned} V_{out} &= A(V_{in} - V_A) = A(V_{in} - \beta V_{out}) \\ V_{out}(1 + \beta A) &= A V_{in} \\ \frac{V_{out}}{V_{in}} &= \frac{A}{1 + \beta A} \end{aligned}$$

(d)

$$Gain = \frac{1}{\beta} = 100$$

(e) For

$$A = 10^9, Gain = 99.99999 \text{ (} 10^{-5} \% \text{ error)}$$

Instrumentation

$$A = 10^6, \text{Gain} = 99.99 \text{ (0.01\% error)}$$

$$A = 10^3, \text{Gain} = 90.9 \text{ (10\% error)}$$

Most op-amps have a gain in the region 10^6 to 10^9 so – as long as we are not using a very large closed-loop gain – the equations derived from the ideal amp approximation are valid.

4.

(a) From the handout for the Instrumentation Amplifier we have

$$V_{out} = (V_2 - V_1) \left(\frac{R_4}{R_3} \right) \left(1 + 2 \left(\frac{R_1}{R_2} \right) \right)$$

Which in this case simplifies to

$$V_{out} = (V_2 - V_1) \left(1 + \frac{50}{1} \right)$$

Hence we find that the differential gain $A_d = 51$.

(b) These equations apply to the ‘perfect’ Instrumentation Amplifier, i.e. they do not take into account any additional output due to common-mode gain. If we define

$$V_d = V_2 - V_1 = \text{differential input}$$

$$V_{cm} = \frac{1}{2}(V_2 + V_1) = \text{common-mode input}$$

$$A_{cm} = \text{common-mode gain}$$

Then

$$V_{out} = A_d V_d + A_{cm} V_{cm}$$

Since

$$CMRR = 20 \log \left(\frac{A_d}{A_{cm}} \right) = 120$$

$$A_{cm} = \frac{51}{10^6}$$

For $V_d = 0.1V$ and $V_{cm} = 0V$, $V_{out} = 51 \times 0.1 = 5.1V$

For $V_d = 0.1V$ and $V_{cm} = 10V$, $V_{out} = 51 \times 0.1 + 51 \times 10^{-6} \times 10 = 5.10005V$

This is a deviation of just 0.001% compared to the zero common-mode input case, from which we can conclude that the INA114 has excellent common-mode rejection!

5.

We need the trig identity

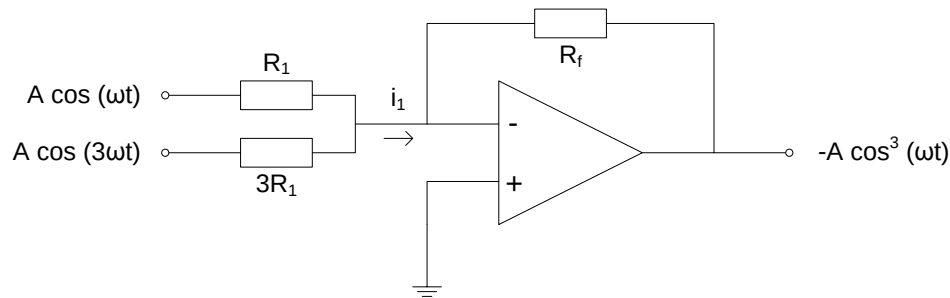
$$\cos^3 x = \frac{1}{4}(3 \cos x + \cos 3x).$$

Let's assume that we set the two signal generators to produce $A \cos \omega t$ and $A \cos 3\omega t$ (A and ω are arbitrary). The desired output is $-A \cos^3 \omega t = -\frac{1}{4}A(\cos 3\omega t + 3 \cos \omega t)$. One way to achieve this would be to use three op-amps:

Instrumentation

- One non-inverting amp to multiply $A \cos \omega t$ by 3.
- One inverting summing amp to add $A \cos 3\omega t$ to the result of the first amp and multiply result by -1.
- One non-inverting amp to scale the result of this by $\frac{1}{4}$.

However, consider the circuit below which should do the whole process in one step



From lectures we know

$$\frac{V_{out}}{R_f} = -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2}\right)$$

So the current in each resistor weights the proportion of each signal in the addition. Hence $R_2 = 3R_1$ will weight the current i_1 in the required ratio. R_f will also be some factor of R_1 so setting $R_f = nR_1$ we just need to find n . Substituting into the above equation gives

$$-\frac{1}{4nR_1} A(\cos 3\omega t + 3 \cos \omega t) = -A \left(\frac{\cos \omega t}{R_1} + \frac{\cos 3\omega t}{3R_1} \right)$$

Which simplifies to

$$n = \frac{3}{4}$$